

familiar with FPGA design flows. Xilinx has been actively working on the UltraFast methodology to shorten this potential initial learning curve by providing an implementation framework and best-in-class examples."

"Moreover, the security (related to code modification, corruption, etc.) that is inherently supported by ASICs or MCUs is not easily achievable with FPGAs; thereby adding another question mark in their abundant use," explains T. Anand, managing director, Knewron.

You also have to learn to make good use of the space available. "Hence the programming should be tight, and the designer should have knowledge of how the FPGA works and how to save space using techniques specific to FPGA. Secondly, certain standard given things which we take for granted on Windows, will not work directly on the FPGA like floating point or direct memory access transfer," adds Vinod Mathews.

Sanchit Bhatia, application engineer, Agilent Technologies India Pvt Ltd says that most T&M equipment like oscilloscopes, logic analyzers, network analyzers need not move to FPGA-based systems as they need to be easy to use and don't require real-time processing. Instruments like digitizers and modular instruments can go the FPGA way since real-time processing is required.

"FPGA-based T&M equipment are in general complex to use and require the users to learn special tools/toolkits to program their on-board FPGAs. Thus FPGA-based T&M equipment are mainly for advanced users who are already power users of T&M equipment. The learning curve associated with FPGA-based equipment are also higher," he adds.

Future trends

For test and measurement vendors, an emerging trend that Stephen says Aeroflex is a keen proponent of, is to build common test equipment platforms that can be used for multiple applications. "This not only saves

R&D cost by avoiding duplication of design effort but more importantly shortens the design cycle—critical for staying abreast of market needs. For example, our Common Platform Architecture has enabled delivering multiple products to different customers addressing different test needs all based on the same core hardware and software. These have included LTE mobile phone testers, WLAN and mobile phone production test equipment, military radio testers and avionics test equipment." Apart from giving customer access to test equipment early, another major customer advantage is the protection of the investment as technology matures. In the past, a new technology standard usually meant designers and manufacturers would need to buy new test equipment completely. Architectures such as the Common Platform using FPGAs and DSPs mean that certain standard updates can be delivered via software only and more significant changes are usually now just an upgrade rather than a complete new purchase.

On the other hand, user-programmable COTS FPGAs are also emerging as a likely platform for next-generation embedded test and measurement solutions, which can be employed during design, development, manufacturing and in the field following product launch.

Chinmay explains that as systems grow even more complex, tools enabling a platform-based approach for system design, and software-defined instrumentation are becoming more popular. This is fuelling the growth of an eco-system around the platform, where many domain experts are creating domain-specific IPs, and sharing it through channels like LabVIEW tools network. With FPGA personalities easily available because of the ecosystem, it is becoming possible for an average test engineer to make a generic test system to test multiple products with the same hardware setup. This is driving down the cost and increasing productivity. ●

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Tools For Learning



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